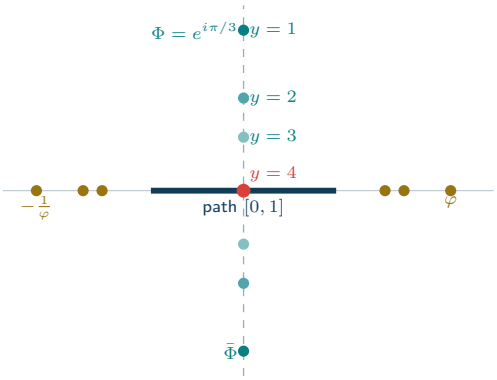
COMBINATORICS ON ONE PAGE

Catalan numbers and the golden ratio

Reciprocal Catalan sums reduce to an integral whose denominator is a golden quadratic, giving $\ln\varphi$ at $y=-1$ and $\pi\sqrt{3}$ at $y=1$.

1 / THE RULE AND THE PICTURE

The sum becomes an integral over $[0,1]$ with a quadratic in the denominator



roots of $1-y\,t(1-t)$ in the t-plane:
teal $y=1,2,3$, gold $y=-1,-2,-3$

The Catalan numbers $C_n=\frac{1}{n+1}\binom{2n}{n}=1,1,2,5,14,42,\ldots$ have reciprocals that a Beta integral turns into

$$\boxed{\tfrac{1}{C_n}=(n+1)(2n+1)\int_0^1\big[t(1-t)\big]^n\,dt}$$

Summing against y^n and using $\sum(n+1)(2n+1)u^n=\frac{1+3u}{(1-u)^3}$ with $u=y\,t(1-t)$ gives

$$\sum_{n\ge0}\frac{y^n}{C_n}=\int_0^1\frac{1+3u}{(1-u)^3}\,dt.$$

The integrand has poles where $1-y\,t(1-t)=0$, that is $t^2-t+\frac{1}{y}=0$, and the picture follows these roots as y changes.

2 / WHERE THE ROOTS GO

The golden ratio sits at $y=-1$ and its complex companion at $y=+1$

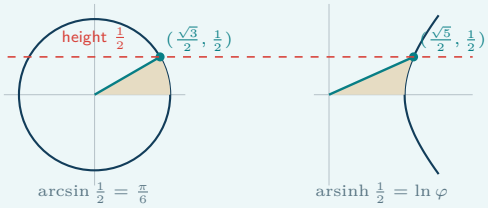
For $0<y<4$ the roots are $t=\frac12\pm\frac{i}{2}\sqrt{\frac{4}{y}-1}$, on the vertical line $\operatorname{Re}t=\frac12$, and they slide down to meet at $t=\frac12$ when $y=4$. There they pinch the path $[0,1]$, so the series has radius 4, matching $C_n^{1/n}\to4$. For $y<0$ the roots are real, $t=\frac12\pm\frac12\sqrt{1+\frac{4}{|y|}}$, outside $[0,1]$. At the two special values the quadratic is a golden one:

$$\boxed{y=-1:\ t^2-t-1=0,\ t=\varphi,-\tfrac{1}{\varphi}\qquad y=+1:\ t^2-t+1=0,\ t=e^{\pm i\pi/3}}$$

the real golden ratio and the sixth roots of unity, which are the twin golden-type polynomials $x^2-x\mp1$.

3 / A ROTATION AND A BOOST

The same height $\frac12$ gives $\frac{\pi}{6}$ on the circle and $\ln\varphi$ on the hyperbola



The simplest integral of this kind shows the pattern, since $\int_0^1\frac{dt}{1-y\,t(1-t)}=\frac{4\arcsin(\sqrt{y}/2)}{\sqrt{y(4-y)}}$ for $0<y<4$. At $y=1$ it is $\frac{4}{\sqrt3}\cdot\frac{\pi}{6}=\frac{2\pi}{3\sqrt3}$ and at $y=-1$, where $\arcsin$ becomes arsinh, it is $\frac{4}{\sqrt5}\ln\varphi$. Both come from the height $\frac12$: on the unit circle $\sin\frac{\pi}{6}=\frac12$ with $\cos\frac{\pi}{6}=\frac{\sqrt3}{2}$, and on the unit hyperbola $\sinh(\ln\varphi)=\frac12$ with $\cosh(\ln\varphi)=\frac{\sqrt5}{2}$. The $\sqrt3$ and the $\sqrt5$ are $\sqrt{4-y}$, and the rotation by 30° and the rapidity $\ln\varphi$ are the circular and hyperbolic versions of one angle.

4 / THE SUMS

Reciprocal Catalan numbers for several y

y	roots of $t^2-t+\frac{1}{y}$	angle or rapidity	$\sum_{n\ge0}\frac{y^n}{C_n}$
1	$e^{\pm i\pi/3}$, the complex golden ratio	30° $(\arcsin\frac12)$	$2+\frac{4\sqrt3\,\pi}{27}=2.8061$
2	$\frac{1\pm i}{2}$	45°	$5+\frac{3\pi}{2}=9.7124$
3	$\frac12\pm\frac{i}{2\sqrt3}$	60°	$22+8\sqrt3\,\pi=65.5312$
-1	φ and $-\frac{1}{\varphi}$, the golden ratio	rapidity $\ln\varphi$ $(\operatorname{arsinh}\frac12)$	$\frac{14}{25}-\frac{24\ln\varphi}{25\sqrt5}=0.3534$

The alternating series that starts at $n=1$ is $1-\frac12+\frac15-\frac{1}{14}+\cdots=1-\sum_{n\ge0}\frac{(-1)^n}{C_n}=\frac{11}{25}+\frac{24}{25\sqrt5}\ln\varphi=0.6466$, which is why a logarithm of the golden ratio appears in a counting sequence. The terms shrink by a factor tending to $\frac14$, and thirteen terms give six decimals.

ASIDE / TRY THESE

Check $\sum\frac{1}{C_n}=1+1+\frac12+\frac15+\cdots=2.8061$ with ten terms (2.8060). For $y=2$ the roots $\frac{1\pm i}{2}$ are at 45° and the sum contains π alone.

LINKS TO OTHER SHEETS

The golden-type numbers sheet treats $x^2-x\mp1$; the polygon sheets treat $e^{i\pi/3}$; the power series sheet treats $\ln$.

Convention. C_n the Catalan numbers; $\varphi=\frac{1+\sqrt5}{2}$; all closed forms were checked against the series and the integral to many digits.

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